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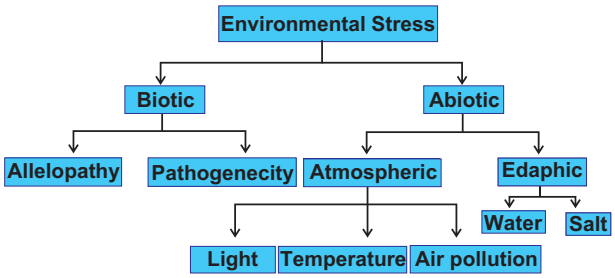


Figure 15.31: Classification of Stress types in plants

competing plants. Biotic environmental stress is also caused due to the activity of man by cutting herbs and trees, twigs for fodders, fuels and agricultural purposes. The biotic stresses caused by bacteria, fungi and nematodes that are ever present in the environment are called potential biotic stresses. These are divided into two types. They are:

i) Allelopathy

An organism producing one or more biochemical substances that greatly influence the germination, growth and reproduction of other organisms is called **Allelopathy**. These biochemicals are known as **allelochemicals**. They are beneficial (positive allelopathic) or detrimental (negative allelopathic). These allelochemicals are obtained from leaf after leaching on the ground and also from roots. The term allelopathy is from Greek words *allelon*-each other and *pathos*-to suffer and first used in 1937 by **Hans Molisch**. Allelopathic effect may occur with weeds on crops and *vice versa* (Figure 15.32).

One of the most famous allelopathic plants is Black walnut (*Juglans nigrum*). The chemical which is present in Black walnut is Juglone and it is a respiratory inhibitor. Solanaceous plants such as tomato, capsicum and eggplant are susceptible to juglone. These plants when exposed to these

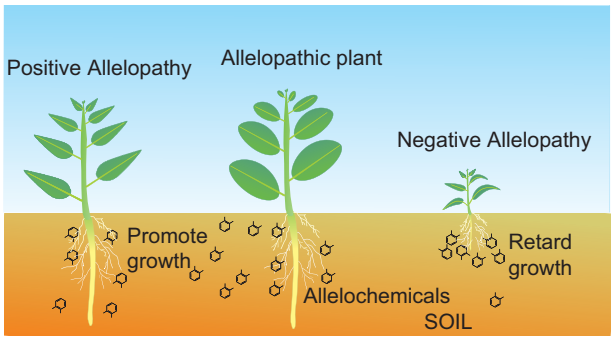


Figure 15.32: Allelopathy in plants

allelochemicals exhibit symptoms such as wilting, chlorosis and death.

Check your grasp! Are all plants allelopathic? Can allelopathic chemicals affect animals and humans?

Tree of heaven (*Ailanthus altissima*) is a recent addition to the list of allelopathic trees. **Ailanthone** an allelochemical extracted from the root of *Ailanthus* acts as potent herbicide. In *Sorghum* plant the allelochemical sorgolone possess allelopathic activity. It is found in root exudates of most *Sorghum* species. Root exudation of maize inhibits the growth of some weeds such as *Chenopodium album* and *Amaranthus retroflexus*. The seed exudates of oat *(Avena fatua)* affect the germination of wheat seedling.

ii. Pathogenecity

The effect of microbes that cause diseases in plants. Example: *Xanthomonas citri*

2. Abiotic Stresses

Abiotic stress may occur due to an atmospheric condition (atmospheric stress) or soil condition (edaphic stress). Atmospheric stresses may occur due to excess and deficient levels of light temperature and air pollutants.